

Toney Zhen

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Education

San Jose State University

2025–2027 (Expected)

M.S. in Artificial Intelligence

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Reinforcement Learning, Intelligent Autonomous Systems

University of California, Irvine

2020–2024

B.S. in Computer Science

GPA: 3.76/4.00

- **Relevant Coursework:** Web Apps, Data Management, Algorithms, Operating Systems, System Design
- **Awards & Activities:** Regents' Scholar; Officer of Chess Club & Pokemon GO Club; member of ACM & Hiking Club

Skills

Programming Languages: Python, Java, JavaScript, C, C++, SQL

AI/ML: PyTorch, torchvision, Scikit-learn, NumPy, Pandas, Matplotlib

Frameworks & Tools: Django, Flask, Node.js, React, jQuery, JUnit, Git, Docker, Kubernetes, AWS, VS Code

Experience

Pan Research Group at UCI

Oct. 2023–Jan. 2025

AI/ML Research Assistant

Irvine, CA

- Trained a PyTorch model on a dataset of **300+** labeled images, achieving **79%** accuracy in grain boundary detection, replacing a slow manual process and freeing researchers to focus on analysis
- Leveraged NumPy, SciPy, and Matplotlib to predict TEM to HRTEM image transformations, improving data quality
- Collaborated weekly with faculty and grad students, incorporating their feedback to refine models and validate results

Federal Geographic Data Committee

Sep. 2023–May 2024

Applied Science Research Intern

Remote

- Developed Python scripts to automate management of spreadsheet data, saving hours of manual effort while ensuring data integrity
- Worked under a remote supervisor and adapted deliverables based on feedback

Projects

Deepfake Image Classifier

Python, Scikit-learn, PyTorch, torchvision

- Fine-tuned pretrained ResNet and EfficientNet CNNs in PyTorch to classify real versus AI-generated images across two datasets, establishing baselines to detect synthetic media
- Developed and tuned ConvNeXt-Tiny, MaxViT, and dual-branch MaxViT models, raising in-distribution validation AUC from **0.84** (EfficientNet CNN baseline) to **0.99**
- Performed cross-dataset evaluation and Grad-CAM analysis, showing performance degraded on unseen data and identifying where models failed to generalize

LLM Safety Guardrails

Python, Scikit-learn, NumPy, Pandas, Matplotlib

- Built a sensitive-data (PII) input guardrail to block private information before it reaches an LLM, training and comparing multiple classifiers on a **300,000**-example PII dataset
- Raised the top model to **92.9%** accuracy, **97.6%** recall, and **96.1%** F1 by replacing TF-IDF features with text embeddings and PCA, improving detection while cutting training runtime by almost half
- Coordinated with a four-person team, building four complementary input/output guardrails, aligning on shared precision and recall targets so results stayed comparable across models

FabFlix MovieMart

JavaScript, jQuery, Java, Jakarta Servlet, MySQL

- Built a full-stack e-commerce web app over a catalogue of **20,000+** movies and **70,000+** stars, with sessions, fuzzy search autocomplete, advanced search options, secure login, and PreparedStatements to prevent SQL injection
- Deployed on AWS EC2 with Docker, a Kubernetes cluster, MySQL replication, and load balancing, sustaining **~9,000** requests/min under Apache JMeter load tests